

Correction

TD - chimie

orga

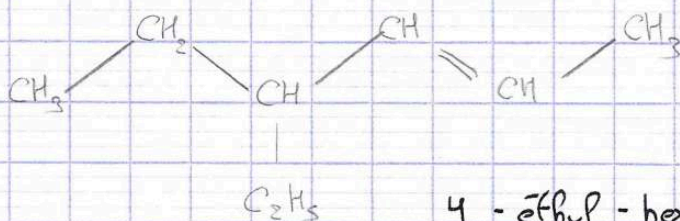
Chimie organique

1

I - Nomenclature

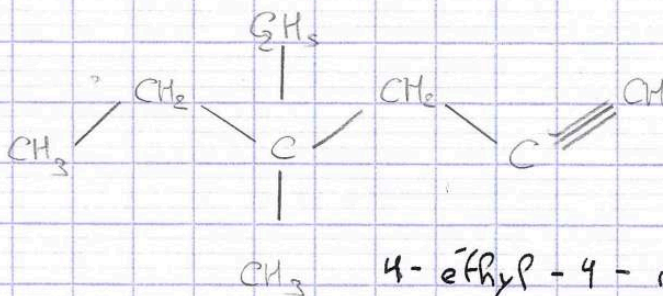
Exercice 1

1)



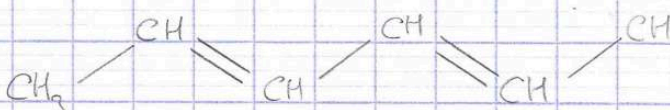
4-éthyl-hex-2-ène

2)



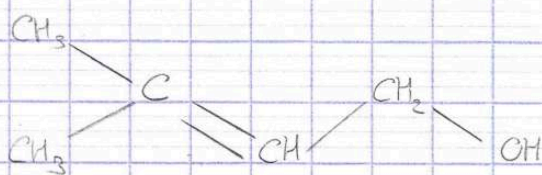
4-éthyl-4-méthyl-1-yne

3)



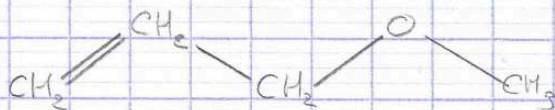
hexa-2,4-diène

4)



3-méthyl-but-2-énol

5)

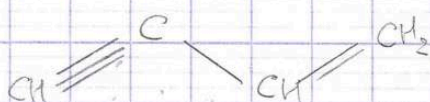


4-oxapent-1-ène

~~3-méthyl-but-2-énol~~

3-méthoxyprop-1-ène

6)

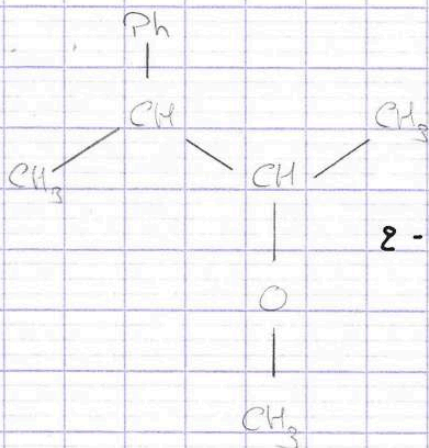


~~4-oxapentène~~

~~3-méthoxypropène~~

butène-3-yne

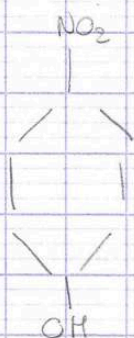
7)



~~butène-3-yne~~

2-méthoxy-3-phénylbutane

8)

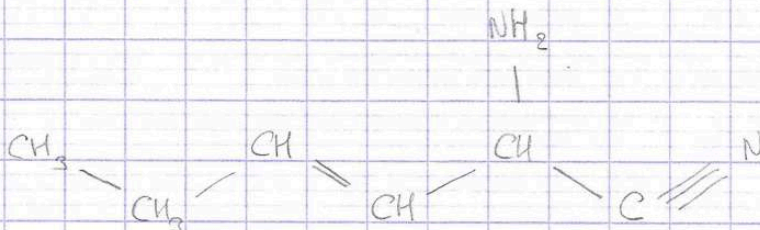


4-nitro phénol
p-nitro phénol

phénol

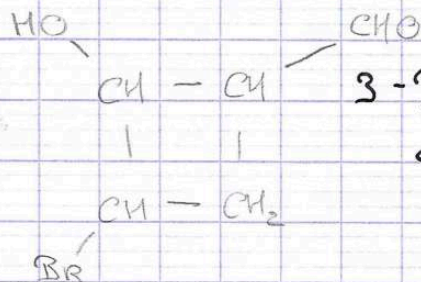


9)



2-amino hex-3-ène-1-nitrile.

10)

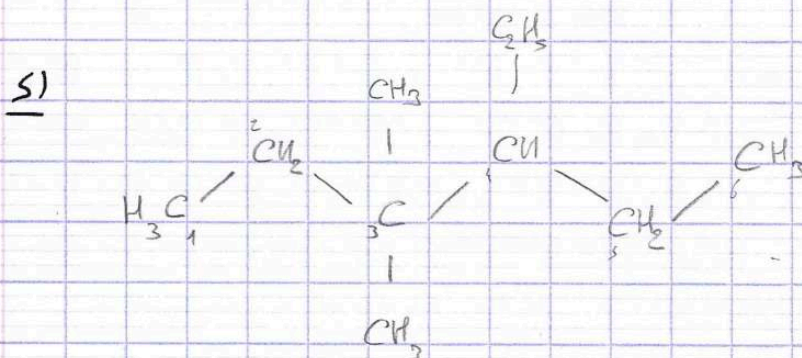
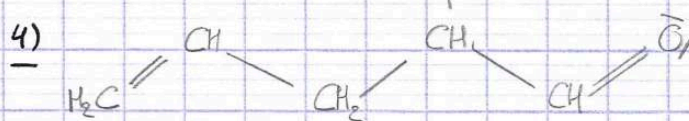
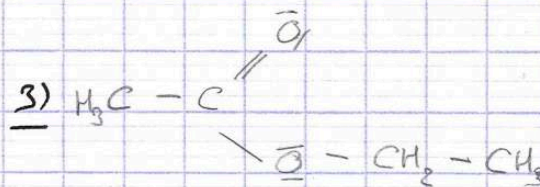
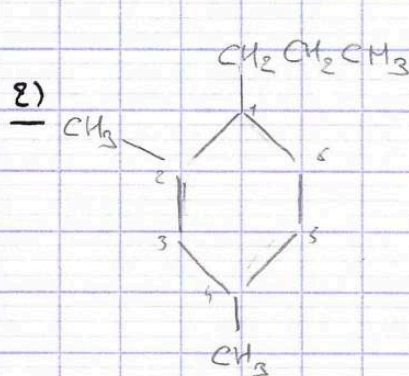
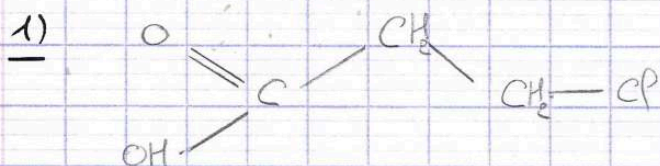


3-Bromo-2-hydroxybutane
carbaldehyde.

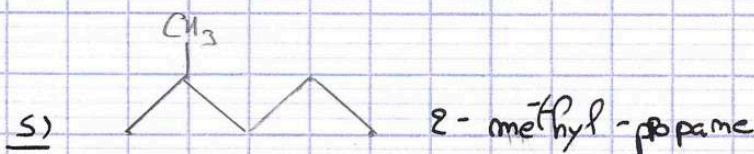
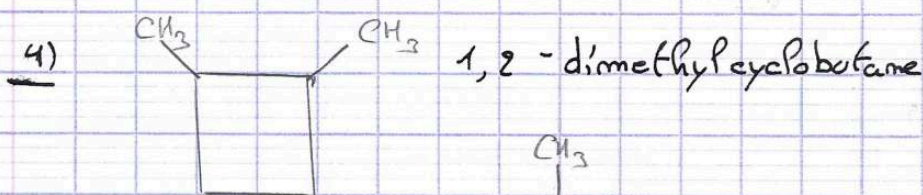
Correction Exercice 2

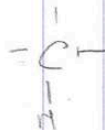
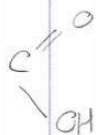
2

ORGA



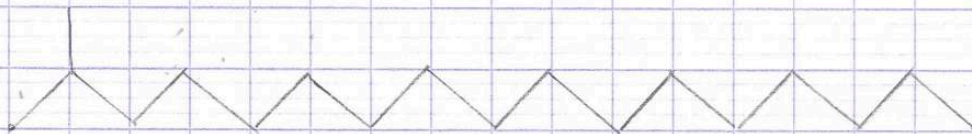
Exercice 3





C: 12
N: 14
O: 16
H: 1

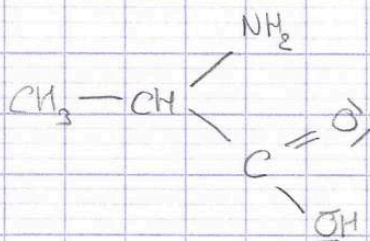
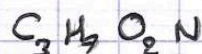
Exercice 4



II - Formule brut et degré d'insaturation

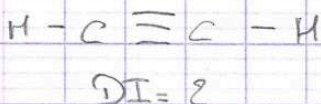
Exercice 1

C : 36 g
H : 7 g
O : 32 g
N : 14 g

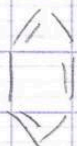


Exercice 2

Degrés d'insaturation (DI) : soit un cycle soit une liaison pi.



DI = 1



DI = 4

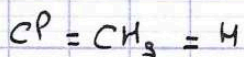
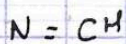
Pour $C_n H_\beta$

On prend l'alcane correspondant : $C_n H_{2n+2}$

$$DI = \frac{2n+2 - \beta}{2}$$

Correction
TD ORG.

Si des atomes autre que C, H :
on les remplace par leur équivalence en
hydrogène :

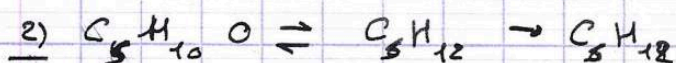


L'exercice 2

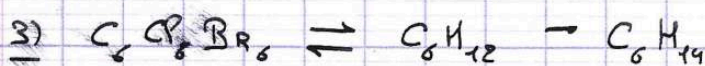
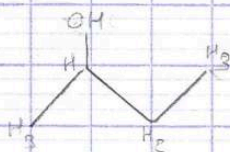
→ - passage alcane correspondant



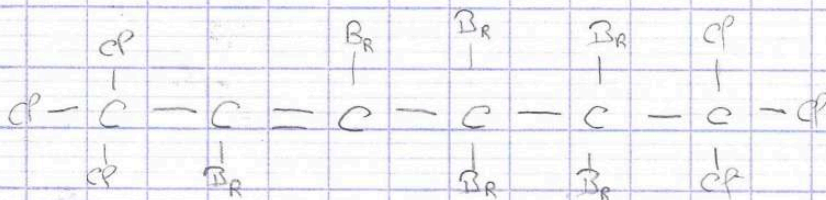
$$DI = \frac{12-6}{2} = 3$$



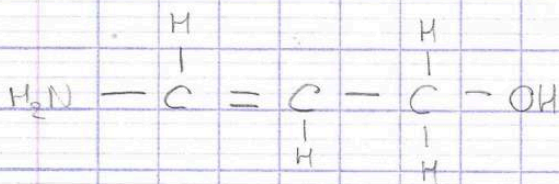
$$DI = \frac{12-12}{2} = 0$$



$$DI = \frac{14-12}{2} = 1$$

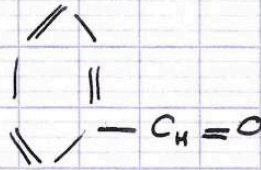
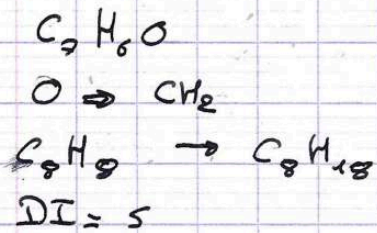


$$DI = \frac{12-10}{2} = 1$$



$$DI = \frac{10-8}{2} = 1$$

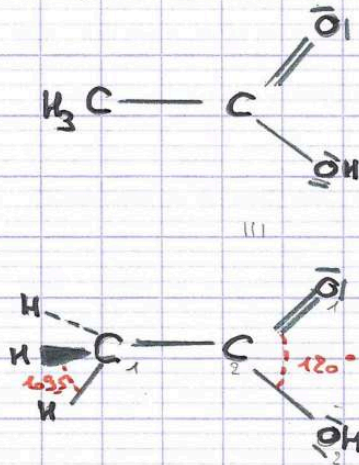
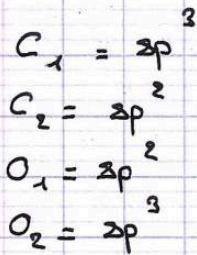
Exercice 3



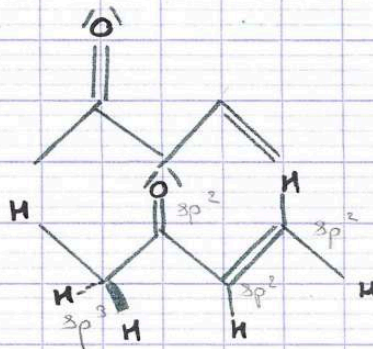
III - Stéréoisomérisme : conformation

Exercice 1

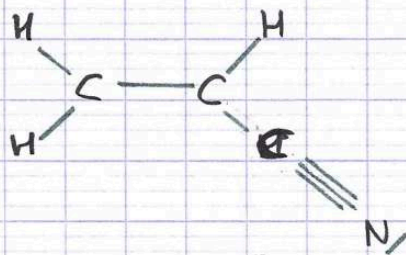
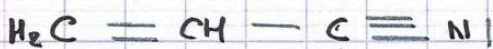
Acide Acétique :



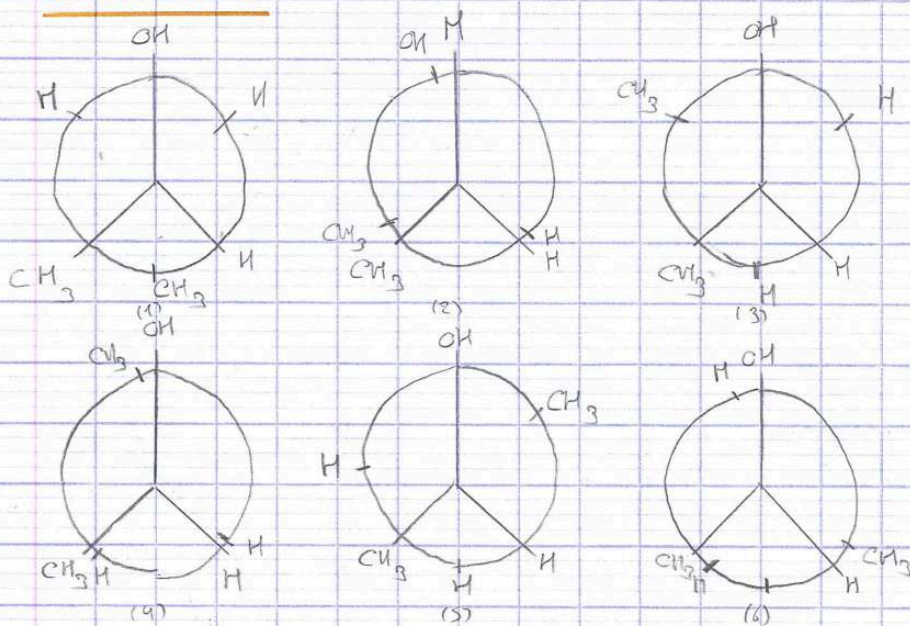
But - 3 - éne - 2 - one



Acrylonitrile



Exercice 2



Décalées

$$\text{OH} - \text{H} = 0,4$$

$$\text{H} - \text{H} = 0$$

$$\text{CH}_3 - \text{H} = 1,2$$

$$\text{CH}_3 - \text{OH} = 2,5$$

$$\text{CH}_3 - \text{CH}_3 = 4,2$$

$$(1) \quad 7,4 \text{ kJ} \cdot \text{mol}^{-1}$$

$$(2) \quad 30,1 \text{ kJ} \cdot \text{mol}^{-1}$$

$$(3) \quad 8,3 \text{ kJ} \cdot \text{mol}^{-1}$$

$$(4) \quad 25,1 \text{ kJ} \cdot \text{mol}^{-1}$$

$$(5) \quad 6,5 \text{ kJ} \cdot \text{mol}^{-1}$$

$$(6) \quad 17,6 \text{ kJ} \cdot \text{mol}^{-1}$$

Éclipsées

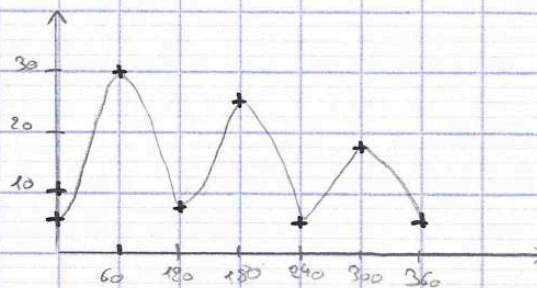
$$\text{CH}_3 - \text{CH}_3 = 20,9$$

$$\text{CH}_3 - \text{OH} = 14,6$$

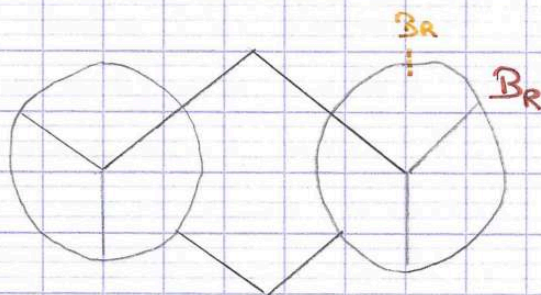
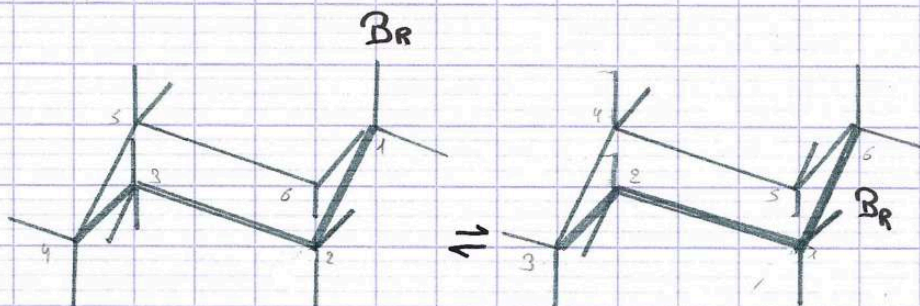
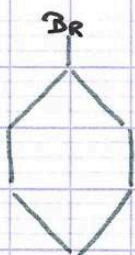
$$\text{CH}_3 - \text{H} = 6,3$$

$$\text{OH} - \text{H} = 5,0$$

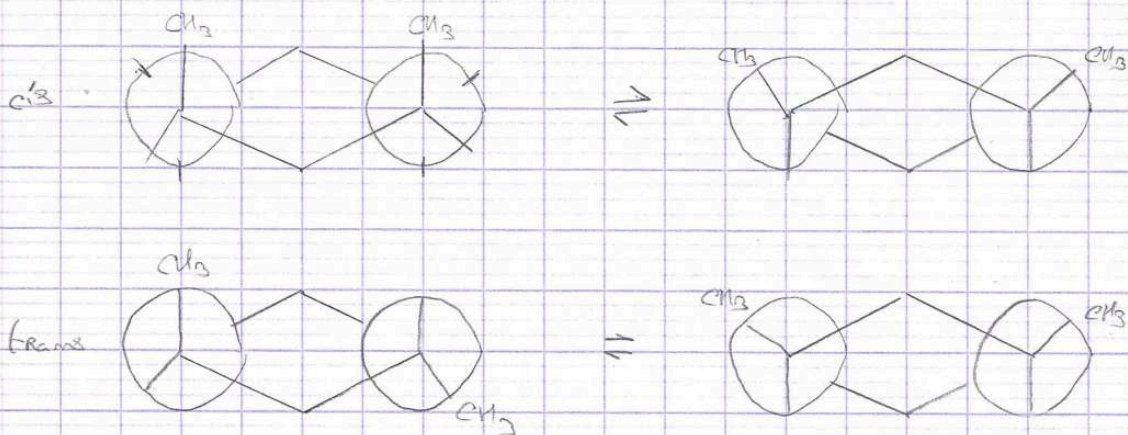
$$\text{H} - \text{H} = 4,2$$



Exercice 3

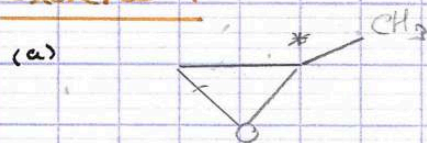


Exercice 4

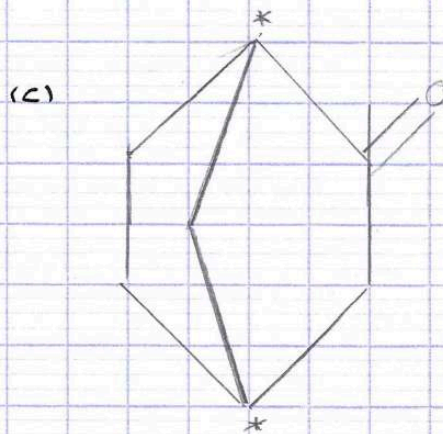
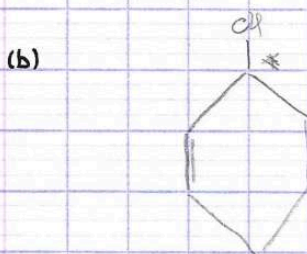


IV - Configuration Énantiomérique

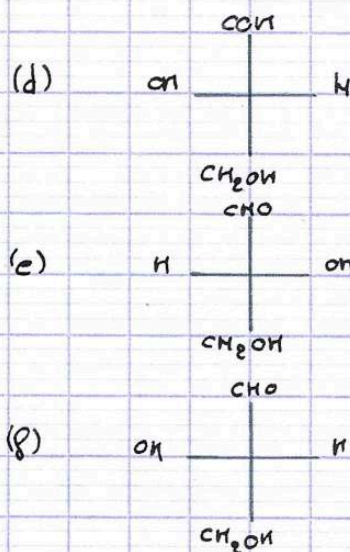
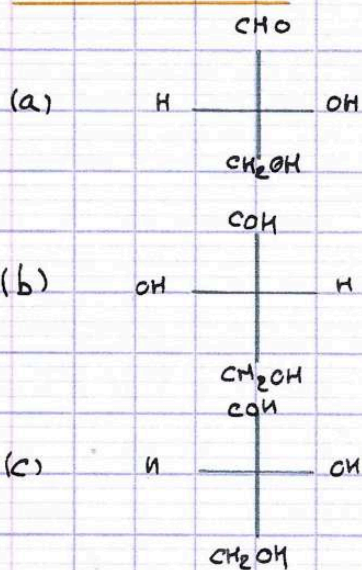
Exercice 1



(d) 1, 6a, 5a, 3a

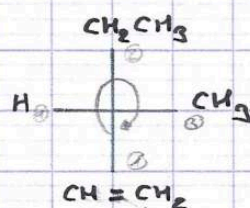
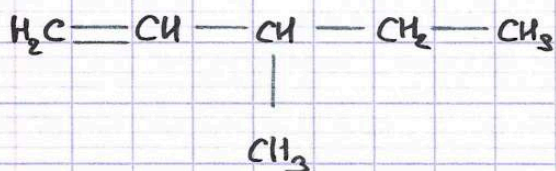
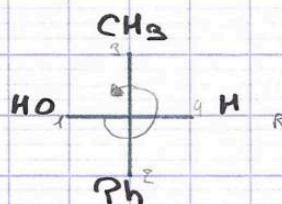
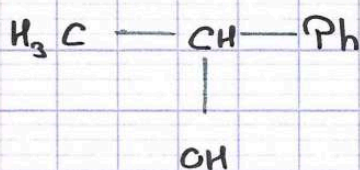
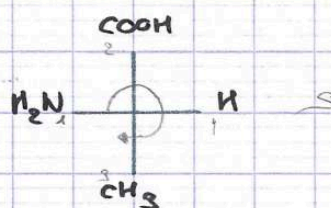
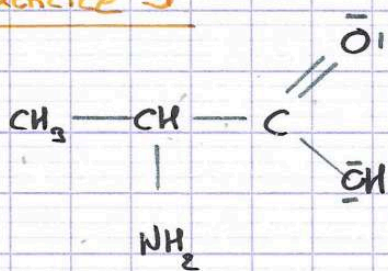


Exercice 3



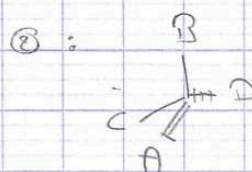
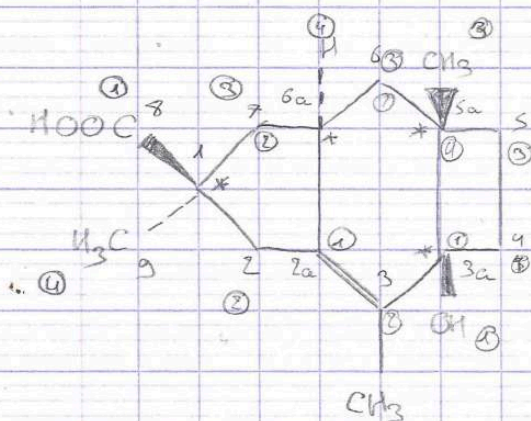
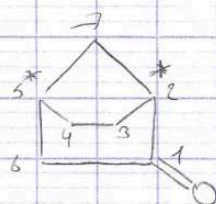
(a) $\Rightarrow$ (c) $\Rightarrow$ (e) $\Rightarrow$ énantiomères
(b) $\Rightarrow$ (d) $\Rightarrow$ (f)

Exercise 3

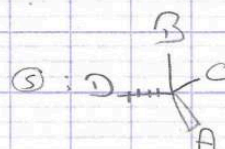


IV - Configuration absolute

Exercise 4



③



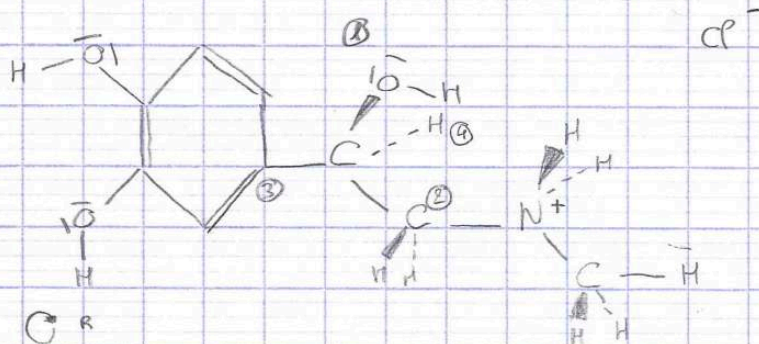
④

- ① : S
- ⑥a : R
- ⑤a : S
- ③a : R

Exercice 5

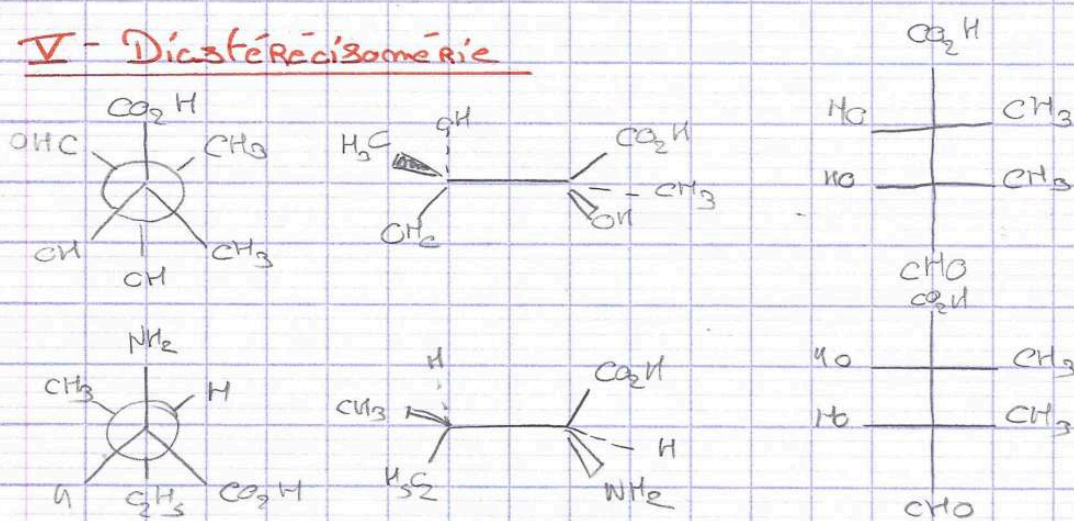
$$[\alpha]_D = \frac{\alpha}{p.c} = +62^\circ$$

Exercice 6

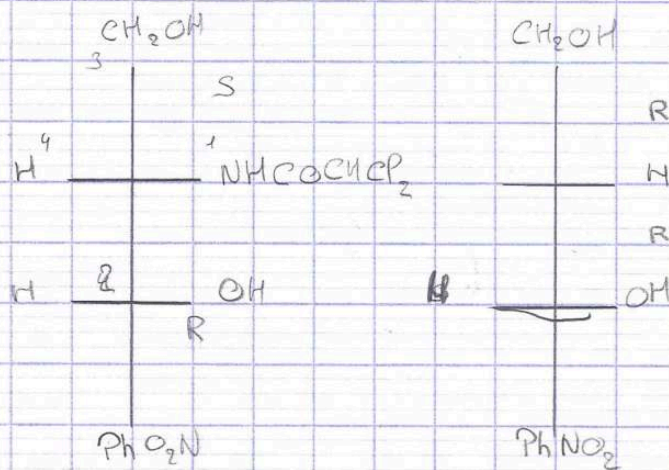


V - Diastéréoisomérisme

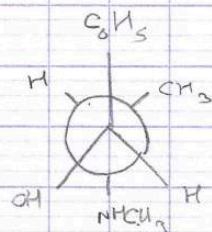
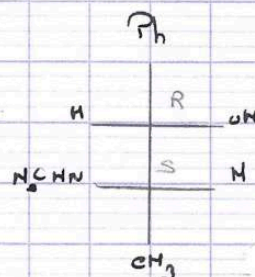
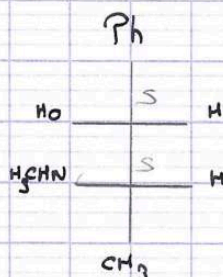
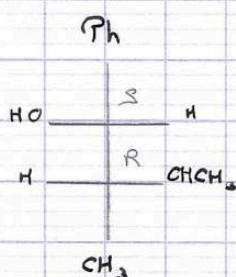
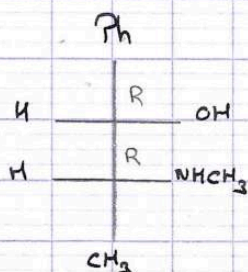
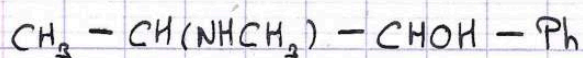
Ex1



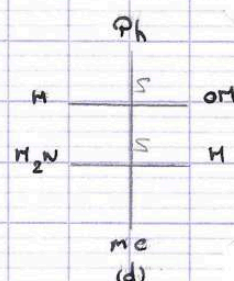
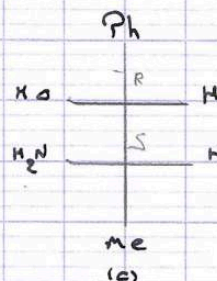
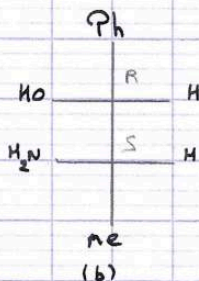
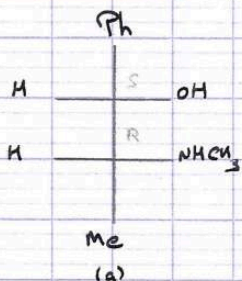
Exercice 3



Exercice 4



Exercice 5



(a) et (b) énantiomères

(b) et (c) identiques

(a) et (c) énantiomères

(a), (b), (c), (d) diastéréoisomères

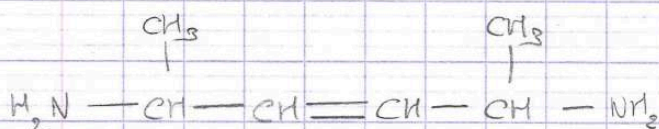
Exercice 6

(a) E

(b) Z

(c) E

(d) Z



Exercice 7

(a) 4

(b) 3 (forme méso)

(c) 4

(d) 8

R R Z
R R E
R S Z
R S E

R R Z
R R E
S S Z
S S E

Alcames : Substitution radicalaire

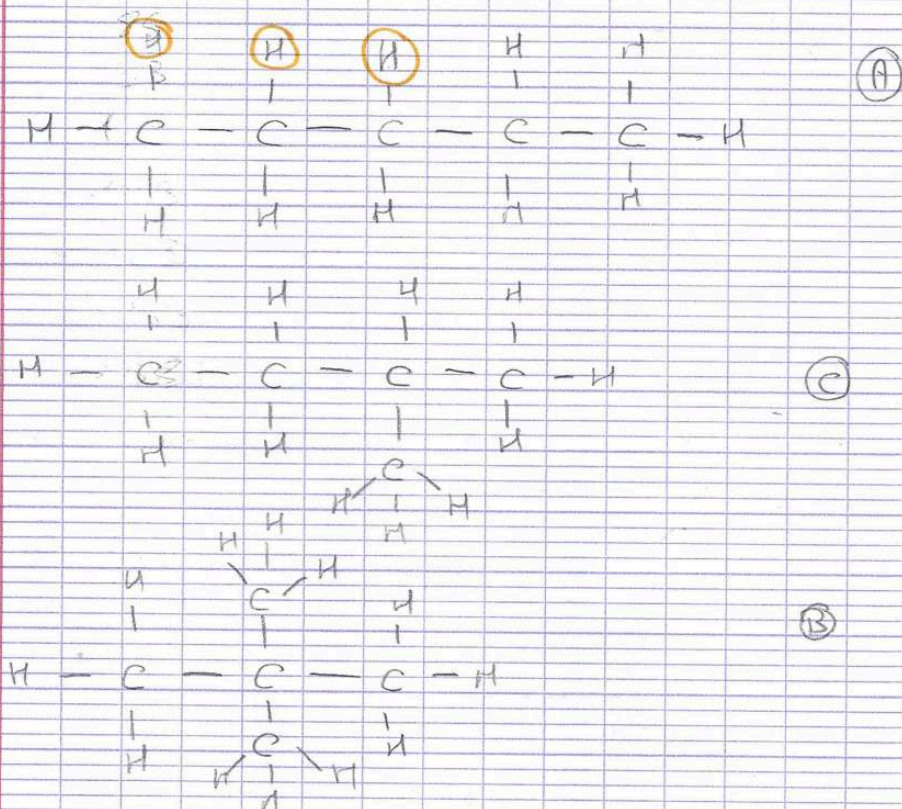
Exercice 1

- A → 3 monochloropentanes
 B → 1 monochloropentane
 C → 4 monochloropentanes

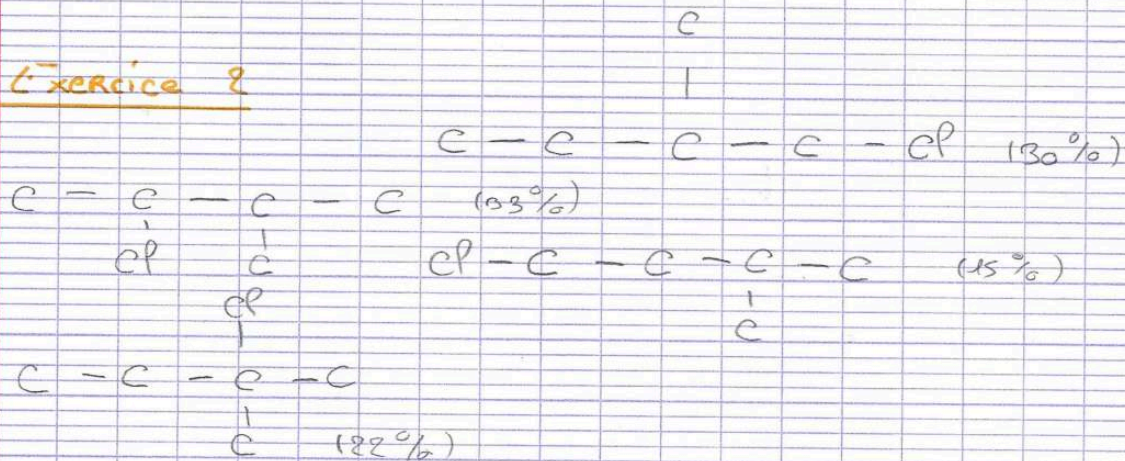
$$\frac{1}{2} + \frac{1}{3} + \frac{1}{8}$$

$$\frac{3 + 2}{6}$$

Pentane C_5H_{12}



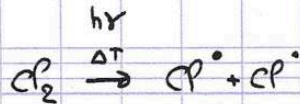
Exercice 2



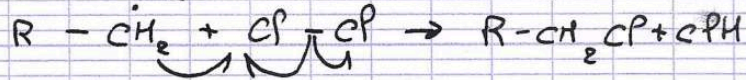
$$15 + 6,4 + 4,6$$

$$21 \quad 26$$

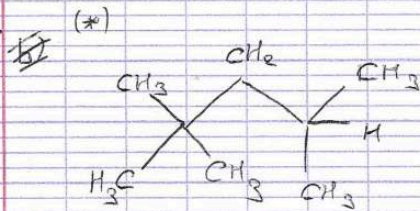
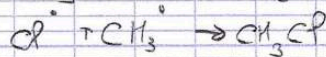
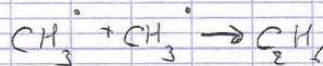
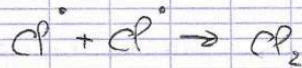
Initiation



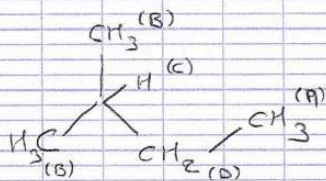
Propagation



Termination



On regarde si les H
sont primaires - secondaires
- tertiaires



$$25\% \text{ A} \quad 3/12 \quad (25\%) \rightarrow 0,6 \rightarrow 1$$

$$50\% \text{ B} \quad 6/12 \quad (50\%) \rightarrow 0,6 \rightarrow 1$$

$$8\% \text{ C} \quad 1/12 \quad (8\%) \rightarrow 2,75 \rightarrow 4,6$$

$$17\% \text{ D} \quad 2/12 \quad (17\%) \rightarrow 0,6 \rightarrow 3,2$$

Pour trouver les pourcentages de (*)

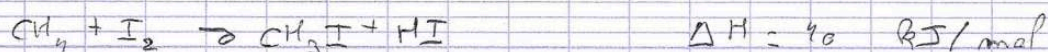
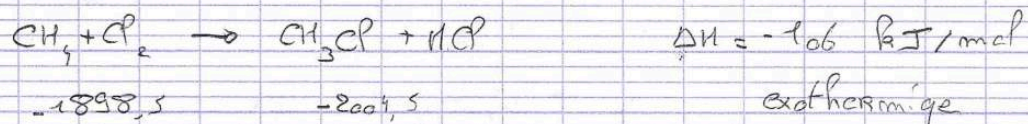
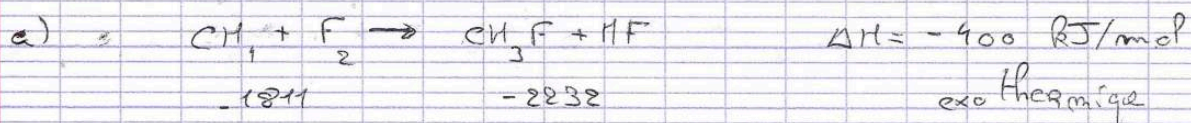
$$\text{Pour C:} \quad \frac{4,6}{15 + 2 \times 3,2 + 4,6} = \frac{4,6}{26} = 17,7\%$$

$$\text{Pour A:} \quad \frac{6}{26} = 23\%$$

$$\text{Pour B:} \quad \frac{3,2 \times 2}{26} = 24,6\%$$

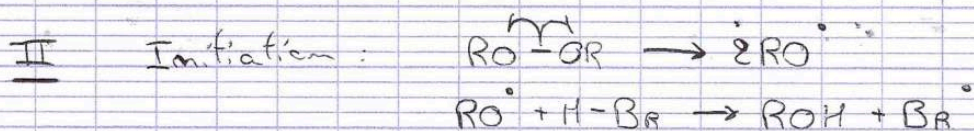
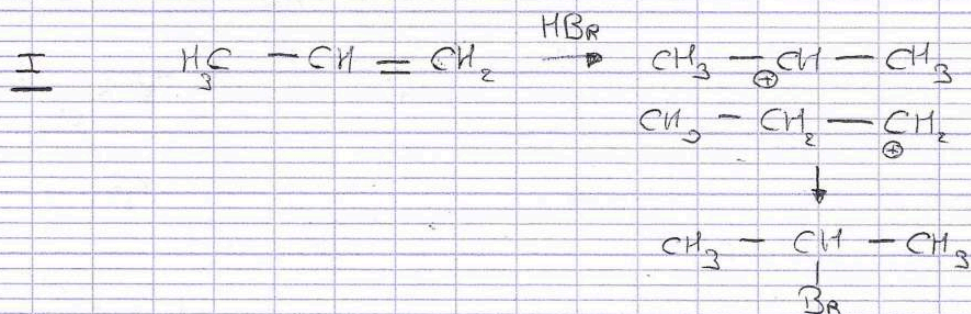
$$\text{Pour D:} \quad \frac{9}{26} = 34,6\%$$

Exercice 3



Alcène - alcynes

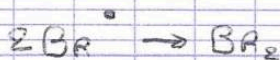
Exercice 1



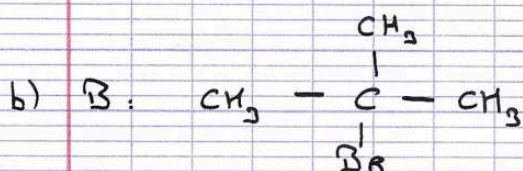
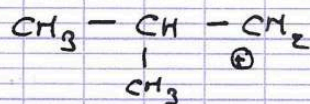
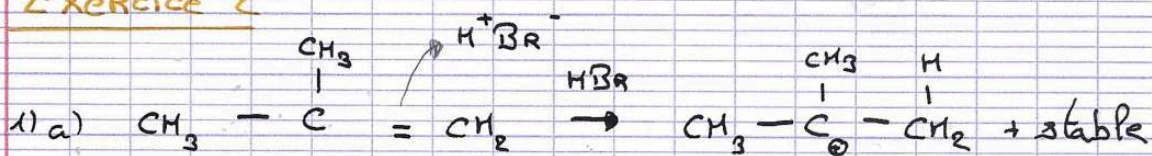
Propagation :



Terminaison

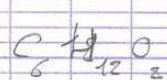


Exercice 2

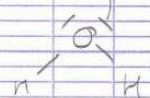
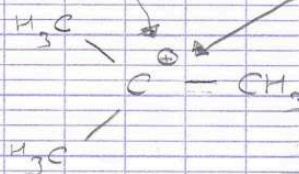
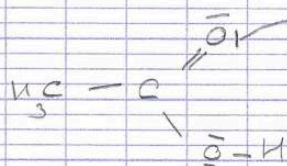
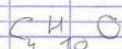
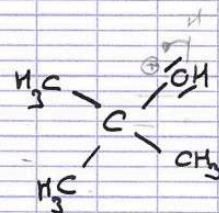
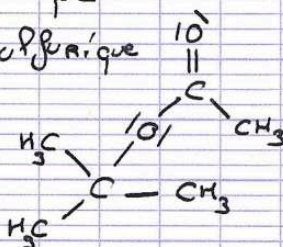
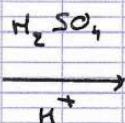
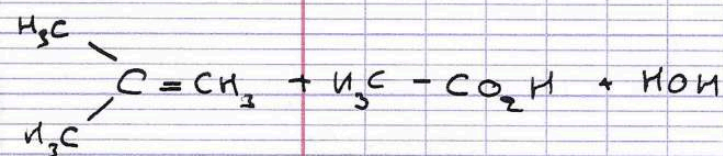


C: $\text{CH}_3 - \text{CO}_2\text{H}$ acide acétique

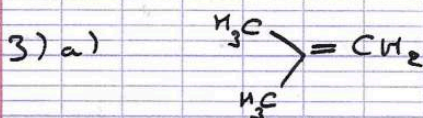
D: H_2SO_4 acide sulfurique



1) a) Par Hydratation



b) On joue sur les proportions d'eau et d'acide acétique



et $\text{H}^+/\text{H}_2\text{O}$ on veut H_2O

Par hydroboration

